

MATHEMATICAL THEORY BEHIND THE SEMANTIC TURING FIELD

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1. INTRODUCTION

The Semantic Turing Field (STF) is a computational model in which words are treated as particles whose interactions are governed by semantic relationships. The motivation is to combine a pretrained semantic representation of language with a dynamical system: a word embedding assigns each word a vector in a high-dimensional semantic space, and pairwise similarity in that space is used to decide whether two particles attract or repel one another. The resulting forces then evolve a separate two-dimensional particle field over time.

It is important to keep two distinct spaces in mind throughout. The *semantic space* is where the embeddings live and where relationships between words are encoded. The *simulation space* is where particles actually move. The purpose of the simulation is not to reproduce the geometry of the embedding space exactly, but rather to provide a dynamic visualization of the semantic relationships that the embedding encodes.

2. SEMANTIC REPRESENTATION

Let the vocabulary consist of N words. Each word i is represented by an embedding vector $\mathbf{v}_i \in \mathbb{R}^d$, where d is the embedding dimension. In the current implementation these vectors are obtained from GloVe.

Semantic similarity between two words is measured by cosine similarity,

$$(1) \quad s_{ij} = \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{\|\mathbf{v}_i\| \|\mathbf{v}_j\|},$$

and the collection of all pairwise similarities forms the matrix $S = (s_{ij}) \in \mathbb{R}^{N \times N}$. This matrix is the sole source of semantic information used by the force model; everything that follows is a consequence of it.

3. SIMULATION SPACE

Each word is associated with a dynamic particle position $\mathbf{x}_i(t) \in \mathbb{R}^2$ and a velocity $\mathbf{u}_i(t) \in \mathbb{R}^2$. The particle position is *not* the original GloVe embedding: the embeddings are first reduced to two dimensions to provide an initial configuration, after which the simulation evolves the particle positions dynamically. The state of the field at time t is therefore the collection

$$\mathcal{S}(t) = \{\mathbf{x}_i(t), \mathbf{u}_i(t)\}_{i=1}^N.$$

Throughout a normal simulation the semantic information is held fixed, while the particle state evolves.

4. THE PAIRWISE FORCE MODEL

The fundamental interaction between two particles is determined by their semantic similarity. Writing $\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i$ for the displacement from particle i toward particle j , the STF pairwise force is

$$(2) \quad \boxed{\mathbf{F}_{ij} = \alpha (s_{ij} - \beta) \mathbf{r}_{ij}},$$

where s_{ij} is the semantic similarity of (1), $\alpha > 0$ controls the overall interaction strength, β is a similarity threshold separating attraction from repulsion, and $\mathbf{r}_{ij}$ gives the spatial direction from i toward j . The total pairwise force acting on particle i is the sum over all other particles,

$$(3) \quad \mathbf{F}_i^{\text{semantic}} = \sum_{j \neq i} \mathbf{F}_{ij}.$$

4.1. Attraction and Repulsion. The qualitative behaviour of the interaction is controlled entirely by the sign of $s_{ij} - \beta$. If $s_{ij} > \beta$ then the coefficient is positive and the force points toward particle j , so the interaction is attractive. If $s_{ij} < \beta$ then the coefficient is negative and the force reverses direction, so the interaction is repulsive. When $s_{ij} = \beta$ the pairwise interaction vanishes. This is summarized in Table 1.

Condition	Interaction
$s_{ij} > \beta$	Attraction
$s_{ij} = \beta$	No pairwise interaction
$s_{ij} < \beta$	Repulsion

TABLE 1. The sign of $s_{ij} - \beta$ determines the qualitative interaction.

The threshold β is therefore not merely a scaling parameter: it determines which semantic relationships are treated as cohesive and which as separating.

4.2. Why Both Attraction and Repulsion Are Needed. It is worth pausing to see why the signed interaction is essential. If every pair of particles attracted one another, the field would collapse into a small region. If every pair repelled, the particles would drift apart without forming any recognizable structure. The signed force provides both local cohesion and global separation at once. The resulting organization is an emergent property of many simultaneous pairwise interactions rather than the output of any explicit clustering procedure.

5. RELATIONSHIP TO SPRING SYSTEMS

The pairwise force in (2) is structurally similar to a spring or graph-based force, because the interaction acts along the line connecting the two particles. A conventional Hookean spring has the form $\mathbf{F} = -k(\mathbf{x}_i - \mathbf{x}_j)$, where the spatial displacement alone determines the magnitude of the force. In STF, by contrast, the spatial displacement determines only the direction, while semantic similarity determines the strength and sign:

spatial displacement $\rightarrow$ direction, semantic similarity $\rightarrow$ interaction strength and sign.

The model can therefore be viewed as a semantic analogue of a force-directed system. The important distinction is that STF does not treat spatial distance itself as the primary source of semantic information; the semantic relationship originates in the embedding space.

6. ENERGY INTERPRETATION

The pairwise interaction admits an energy interpretation. For a fixed similarity value s_{ij} , define the pairwise potential

$$(4) \quad E_{ij} = \frac{\alpha}{2} (s_{ij} - \beta) \|\mathbf{x}_i - \mathbf{x}_j\|^2.$$

Taking the negative gradient with respect to $\mathbf{x}_i$ gives

$$-\nabla_{\mathbf{x}_i} E_{ij} = \alpha (s_{ij} - \beta) (\mathbf{x}_j - \mathbf{x}_i),$$

which is exactly the STF pairwise force (2), so that $\mathbf{F}_{ij} = -\nabla_{\mathbf{x}_i} E_{ij}$.

The interpretation of this potential depends on the sign of $s_{ij} - \beta$. When $s_{ij} > \beta$ the coefficient of the quadratic is positive, so the energy is minimized when the two particles are close. When $s_{ij} < \beta$ the coefficient is negative, so increasing separation lowers the pairwise energy; in an unconstrained system this would encourage particles to move arbitrarily far apart. This is one reason the actual simulation requires additional mechanisms such as boundary forces and damping. The energy interpretation should therefore be understood as a description of the pairwise semantic interaction, not as a claim that the full implemented system is pure gradient descent on a single global energy function.

7. DISCRETE-TIME DYNAMICS

The simulation evolves according to a simple discrete-time integration scheme. The total force on particle i is

$$(5) \quad \mathbf{F}_i(t) = \mathbf{F}_i^{\text{semantic}} + \mathbf{F}_i^{\text{gravity}} + \mathbf{F}_i^{\text{boundary}},$$

and the velocity and position are updated by

$$(6) \quad \mathbf{u}_i(t + \Delta t) = \mathbf{u}_i(t) + \mathbf{F}_i(t) \Delta t,$$

$$(7) \quad \mathbf{x}_i(t + \Delta t) = \mathbf{x}_i(t) + \mathbf{u}_i(t + \Delta t) \Delta t.$$

Damping is applied after the integration step, $\mathbf{u}_i \leftarrow \gamma \mathbf{u}_i$, where γ is the damping parameter. The implementation also limits the maximum particle speed in order to maintain numerical stability and visual control.

8. BOUNDARY FORCES

The semantic interaction alone does not constrain particles to remain within the visible simulation region, so a boundary force $\mathbf{F}_i^{\text{boundary}}$ is added to the total force (5). Boundary forces should be interpreted as simulation constraints rather than as semantic relationships.

9. K-MEANS CLUSTERING

K-Means is used as a complementary structural analysis of the embedding space. Given the word embeddings $\{\mathbf{v}_1, \dots, \mathbf{v}_N\}$, K-Means partitions them into K clusters by minimizing

$$\sum_{k=1}^K \sum_{\mathbf{v}_i \in C_k} \|\mathbf{v}_i - \boldsymbol{\mu}_k\|^2,$$

where $\boldsymbol{\mu}_k$ is the centroid of cluster C_k . The resulting cluster assignment for word i is denoted $c_i \in \{1, \dots, K\}$.

K-Means does not determine the physical force between particles. Cosine similarity drives the semantic interaction, while K-Means supplies cluster assignments for visualization. This distinction matters because a pair of words can belong to the same K-Means cluster without necessarily having a strong direct interaction under the STF force model. K-Means therefore provides a coarse semantic grouping, whereas the force field represents continuous pairwise relationships.

10. GRAVITY WAVE

The gravity wave introduces an external semantic perturbation into the field. Suppose a user enters a sentence q . After preprocessing, the sentence is represented by the mean embedding

$$\mathbf{q} = \frac{1}{|V_q|} \sum_{w \in V_q} \mathbf{v}_w,$$

where V_q is the set of valid vocabulary words appearing in the sentence. For every vocabulary word i we compute the similarity $s_{iq} = \text{cosine_similarity}(\mathbf{v}_i, \mathbf{q})$, and the gravity-wave force is activated for sufficiently relevant words. A simplified representation of this force is

$$(8) \quad \mathbf{F}_i^{\text{gravity}} = -\lambda \max(s_{iq} - \tau, 0) \mathbf{x}_i,$$

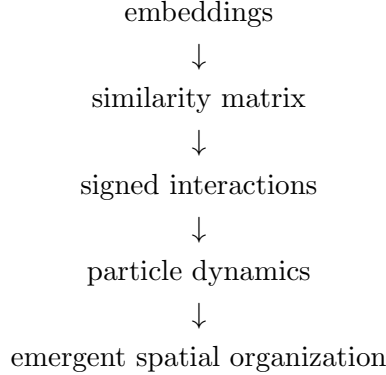
where λ controls the strength of the external influence, τ is the relevance threshold, and s_{iq} measures similarity between word i and the input sentence. Because the force points toward the origin, words semantically related to the input sentence are drawn toward the center of the field. The gravity wave thus acts as a semantic query mechanism:

sentence $\rightarrow$ sentence embedding $\rightarrow$ word-sentence similarity $\rightarrow$ external force $\rightarrow$ field response.

Unlike the pairwise semantic interaction, the gravity wave is externally triggered and temporary.

11. EMERGENT STRUCTURE

No explicit rule instructs the simulation to form semantic communities; structure emerges from the interaction of many particles. The overall process can be summarized as



Semantically related words tend to experience attractive interactions when their similarity exceeds the threshold, while less-related words experience repulsive interactions. The resulting spatial organization is therefore a dynamical representation of the semantic relationships encoded by the embedding model.

12. WHAT STF REPRESENTS

STF should be interpreted as a *semantic dynamical visualization* rather than as a replacement for the original embedding model. The embedding model supplies the semantic information; the STF force model turns that information into interactions; the simulation integrates those interactions over time; K-Means provides a coarse structural grouping for visual interpretation; and the renderer exposes the resulting system to the user. The complete conceptual pipeline is

language $\rightarrow$ embedding $\rightarrow$ similarity $\rightarrow$ force field $\rightarrow$ dynamics $\rightarrow$ visualization,

with K-Means providing an additional clustering layer and the gravity wave providing an interactive semantic perturbation.

13. LIMITATIONS OF THE CURRENT MODEL

The current STF formulation is intentionally simple, and several limitations should be recognized. First, the semantic representation depends entirely on the pretrained embedding model, so any biases or limitations in the embeddings are inherited by the field. Second, the two-dimensional simulation space is not itself a faithful representation of the original high-dimensional semantic space; it is a dynamic visualization space. Third, the threshold β is a manually selected parameter, and different values can produce substantially different field structures. Fourth, the current force model does not explicitly include mass, realistic physical units,

or a physically derived potential governing every component of the system. Finally, K-Means provides a useful visualization aid but should not be interpreted as ground-truth semantic categories. These limitations are intentional in the current version: STF is primarily a computational framework for exploring how semantic relationships can be represented dynamically.

14. FUTURE EXTENSIONS

The current formulation provides a foundation for more sophisticated experiments. Natural extensions include alternative embedding models and clustering algorithms; normalized or distance-dependent force magnitudes; additional force components; spatial partitioning for larger vocabularies; quantitative measures of field stability; systematic experiments over α , β , Δt , and the damping parameter; more principled energy formulations; interactive semantic queries; and reproducible simulation and animation generation. The architecture is intentionally modular so that these extensions can be introduced without replacing the entire system.